CBDVENTURES
DATABASE TESTING,
DATA MANAGEMENT AND BACKUP REPORT

1. Introduction

CBDventures is a local business discovery application intended to help people find businesses and services around Richfield Pretoria Campus. The database stores information about users, business listings, categories, reviews, events, business hours and user preferences. This report documents the database checks carried out for the Database Testing, Data Management and Backup role. It records the SQL validation results and the successful export and restoration of a separate test database.

2. Testing environment and scope

Testing was performed using MySQL Workbench with the schema cbdventures_db_test. A separate schema, cbdventures_db_restore, was used for recovery testing to avoid overwriting the original test data. The tests covered database objects, sample record counts, integrity controls, business rules, export and restoration. These checks do not, by themselves, establish that the Java application is connected to or working with this database.

3. Database structure and baseline data

The database structure check confirmed 12 base tables and one view, vw_business_directory. The sample data contained five users, thirteen businesses, four reviews and two events. These counts were used as the baseline for the restoration comparison.

Object or record	Verified count
Base tables	12
Views	1
Users	5
Businesses	13
Reviews	4
Events	2

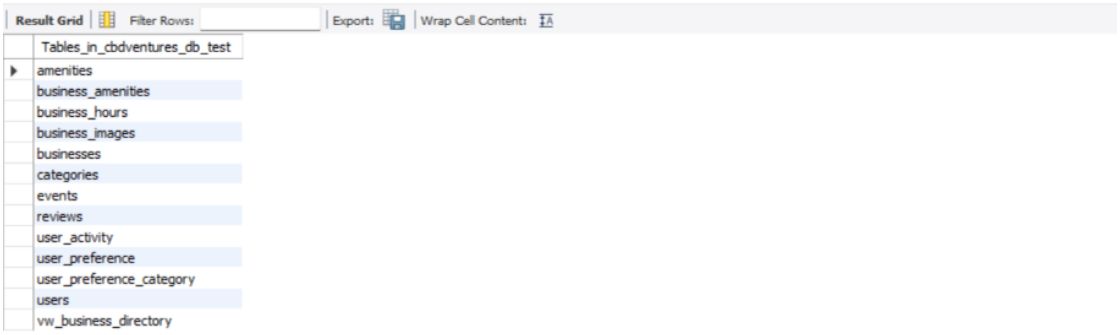


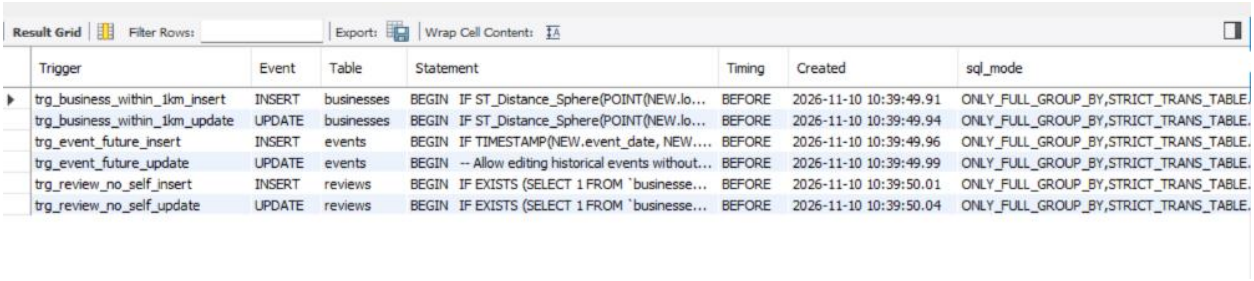
Figure 1. Database tables and business directory view.



Figure 2. Baseline record counts in the original test database.

4. Database triggers and constraints

The SHOW TRIGGERS command returned six installed triggers: two for the 1 km business-location rule, two for future event dates and two preventing owners from reviewing their own businesses. The duplicate-review rule was enforced by a unique key, while review ratings were checked by a CHECK constraint. Trigger presence was verified separately from behaviour tests.



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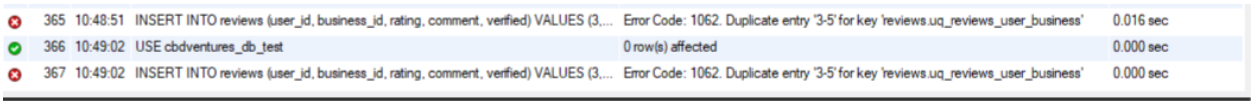
Trigger	Event	Table	Statement	Timing	Created	sql_mode
trg_business_within_1km_insert	INSERT	businesses	BEGIN IF ST_Distance_Sphere(POINT(NEW.lo...	BEFORE	2026-11-10 10:39:49.91	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE.
trg_business_within_1km_update	UPDATE	businesses	BEGIN IF ST_Distance_Sphere(POINT(NEW.lo...	BEFORE	2026-11-10 10:39:49.94	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE.
trg_event_future_insert	INSERT	events	BEGIN IF TIMESTAMP(NEW.event_date, NEW....	BEFORE	2026-11-10 10:39:49.96	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE.
trg_event_future_update	UPDATE	events	BEGIN -- Allow editing historical events without...	BEFORE	2026-11-10 10:39:49.99	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE.
trg_review_no_self_insert	INSERT	reviews	BEGIN IF EXISTS (SELECT 1 FROM `businesse...	BEFORE	2026-11-10 10:39:50.01	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE.
trg_review_no_self_update	UPDATE	reviews	BEGIN IF EXISTS (SELECT 1 FROM `businesse...	BEFORE	2026-11-10 10:39:50.04	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE.

Figure 3. Six installed triggers in the test database.

5. Practical test results

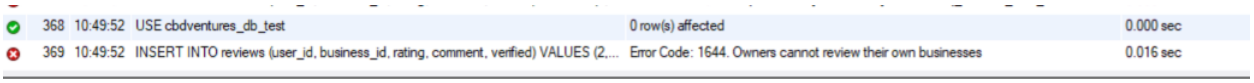
The following tests deliberately attempted invalid changes. For these negative tests, receiving the expected database error counts as a pass because the invalid data was rejected.

Test	Action and expected outcome	Observed result	Status
Duplicate review	Insert a second review for user 3 and business 5; reject duplicate	Error 1062: duplicate entry 3-5	Passed
Owner self-review	Insert review by owner 2 for business 5; reject	Error 1644: owners cannot review their own businesses	Passed
Location restriction	Update business 5 coordinates to Cape Town; reject	Error 1644: business must be within 1 km	Passed
Past event	Insert event dated 1 Jan 2020; reject	Error 1644: event start must be in the future	Passed
Invalid rating	Insert 8-star rating; reject	Error 3819: reviews_chk_1 violated	Passed



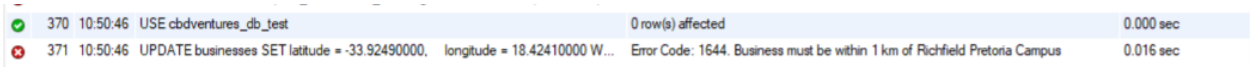
365	10:48:51	INSERT INTO reviews (user_id, business_id, rating, comment, verified) VALUES (3,...	Error Code: 1062. Duplicate entry '3-5' for key 'reviews.uq_reviews_user_business'	0.016 sec
366	10:49:02	USE cbdventures_db_test	0 row(s) affected	0.000 sec
367	10:49:02	INSERT INTO reviews (user_id, business_id, rating, comment, verified) VALUES (3,...	Error Code: 1062. Duplicate entry '3-5' for key 'reviews.uq_reviews_user_business'	0.000 sec

Figure 4. Duplicate review rejected by unique constraint.



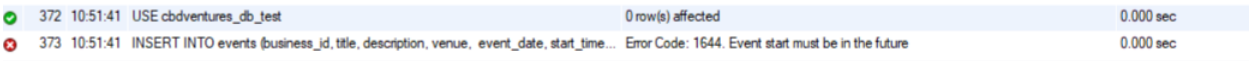
368	10:49:52	USE cbdventures_db_test	0 row(s) affected	0.000 sec
369	10:49:52	INSERT INTO reviews (user_id, business_id, rating, comment, verified) VALUES (2,...	Error Code: 1644. Owners cannot review their own businesses	0.016 sec

Figure 5. Owner self-review rejected by trigger.



370	10:50:46	USE cbdventures_db_test	0 row(s) affected	0.000 sec
371	10:50:46	UPDATE businesses SET latitude = -33.92490000, longitude = 18.42410000 W...	Error Code: 1644. Business must be within 1 km of Richfield Pretoria Campus	0.016 sec

Figure 6. Out-of-area business location rejected.



372	10:51:41	USE cbdventures_db_test	0 row(s) affected	0.000 sec
373	10:51:41	INSERT INTO events (business_id, title, description, venue, event_date, start_time...	Error Code: 1644. Event start must be in the future	0.000 sec

Figure 7. Past event rejected.

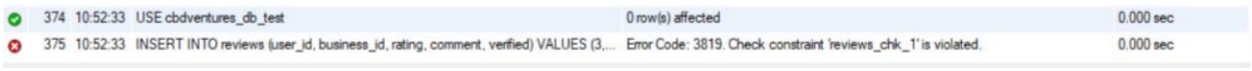


Figure 8. Invalid eight-star rating rejected.

6. Database backup

A MySQL Workbench Data Export was performed for cbdventures_db_test. The export used a self-contained SQL file, Structure and Data, Dump Triggers, Include Create Schema and Create Dump in a Single Transaction. The Export Progress screen reported “Export Completed” and “13 of 13 exported.” This establishes that the export process completed; the restoration test below was used to verify recoverability.

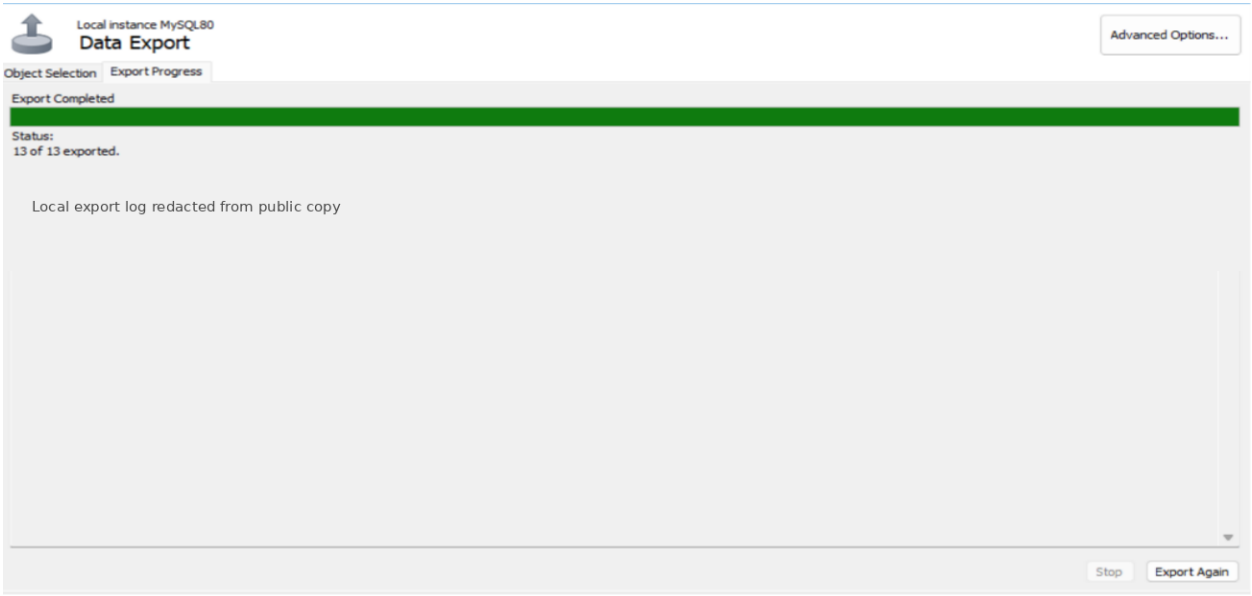


Figure 9. MySQL Workbench export completed

7. Database restoration and verification

The SQL backup was restored into a separate schema named cbdventures_db_restore. The first attempt to select this schema produced MySQL error 1049 because it did not yet exist. After the schema was created and the full restoration script was run, the final commands completed. One collation deprecation warning appeared, but subsequent queries confirmed that the required objects and records were restored.

The restored database returned five users, thirteen businesses, four reviews and two events. A further SHOW TRIGGERS query showed all six triggers. The information_schema.TABLES check confirmed twelve BASE TABLE objects and one VIEW. These results matched the original test database for the checks performed.

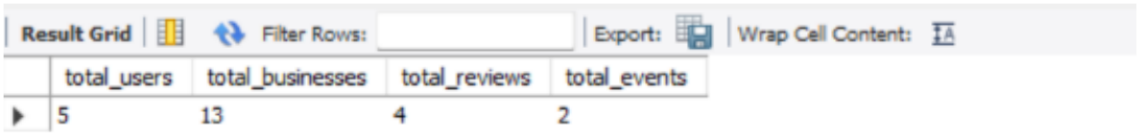


Figure 10. Restored sample record counts match the original.

Trigger	Event	Table	Statement	Timing	Created	sql_mode
trg_business_within_1km_insert	INSERT	businesses	BEGIN IF ST_Distance_Sphere(POINT(NEW.lo...	BEFORE	2026-11-10 11:25:24.54	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABL
trg_business_within_1km_update	UPDATE	businesses	BEGIN IF ST_Distance_Sphere(POINT(NEW.lo...	BEFORE	2026-11-10 11:25:24.70	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABL
trg_event_future_insert	INSERT	events	BEGIN IF TIMESTAMP(NEW.event_date, NEW....	BEFORE	2026-11-10 11:25:25.20	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABL
trg_event_future_update	UPDATE	events	BEGIN -- Allow editing historical events without...	BEFORE	2026-11-10 11:25:25.35	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABL
trg_review_no_self_insert	INSERT	reviews	BEGIN IF EXISTS (SELECT 1 FROM `businesse...	BEFORE	2026-11-10 11:25:25.67	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABL
trg_review_no_self_update	UPDATE	reviews	BEGIN IF EXISTS (SELECT 1 FROM `businesse...	BEFORE	2026-11-10 11:25:25.83	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABL

Figure 11. Six triggers present after restoration.

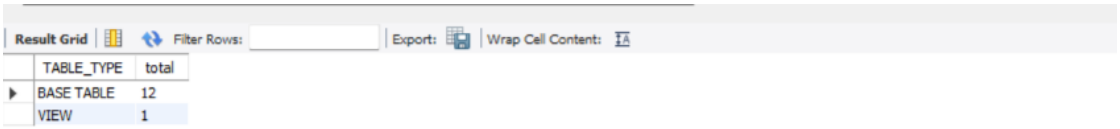


Figure 12. Restored schema contains 12 base tables and 1 view.

8. Challenges and resolutions

Challenge	Resolution / observation
Restoration schema did not exist	Created cbdventures_db_restore and ran the restoration script against the separate schema.
Invalid operations produced errors	Errors were expected in negative tests and showed that the database rules rejected invalid data.
Collation warning during restoration	A deprecation warning appeared; the restored record counts, triggers and object counts were then checked successfully.

9. Conclusion

The database checks documented in this report passed: the expected objects and sample records were present, five invalid-data scenarios were rejected, the SQL backup export completed, and the restored schema matched the original for the verified record counts, triggers and table/view totals. The evidence supports successful database integrity testing and backup recovery for this test environment. Additional application-level integration and any untested rules should be verified separately before production use.

Appendix: Key SQL verification commands

SHOW FULL TABLES;

SHOW TRIGGERS;

SELECT COUNT(*) FROM businesses;

SELECT TABLE_TYPE, COUNT(*) AS total FROM information_schema.TABLES WHERE TABLE_SCHEMA = 'cbdventures_db_restore' GROUP BY TABLE_TYPE;